

TAMIL NADU STATE BOARD - STANDARD 9 SCIENCE (PHYSICS)

CHAPTER 4: GRAVITATION

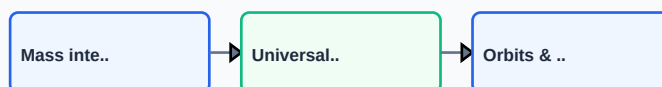
PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 9 English Medium
Subject:	Science (Physics)
Chapter Name:	Gravitation

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

TOPIC ROADMAP: UNIVERSAL LAW OF GRAVITATION



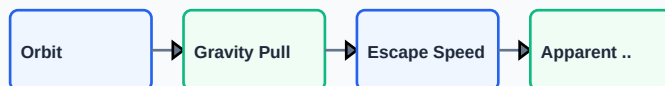
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Universal Law of Gravitation	Every particle in the universe attracts every other particle with a force directly proportional to the product of their masses and inversely proportional to the square of the distance between them.
Acceleration due to Gravity (g)	The acceleration produced in a body due to the gravitational pull of the Earth. Its average value at the surface is 9.8 m/s^2 .
Mass	The quantity of matter contained in a body, which is a scalar quantity and remains constant everywhere in the universe.
Weight	The gravitational force acting on a body towards the center of the Earth, which is a vector quantity and varies with location.
Escape Velocity	The minimum velocity required for a body to escape the gravitational field of a planet and enter outer space.

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

GLOSSARY: GRAVITATIONAL ORBITAL PATHS



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

FORMULA: GRAVITATIONAL FORCE CALCULATION



Essential Formulas, Laws & Identities:

- Newton's Gravitational Force: $F = G * (m_1 * m_2) / d^2$
- Universal Gravitational Constant: $G = 6.674 \times 10^{-11} \text{ N m}^2/\text{kg}^2$
- Relation between g and G : $g = (G * M) / R^2$
- Weight of a body: $W = m * g$
- Apparent Weight in Elevator (moving up with acceleration a): $W = m(g + a)$
- Apparent Weight in Elevator (moving down with acceleration a): $W = m(g - a)$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Calculate the gravitational force between Earth ($M = 6 \times 10^{24} \text{ kg}$) and a 1 kg object on its surface ($R = 6.4 \times 10^6 \text{ m}$).

Step-by-step Solution:

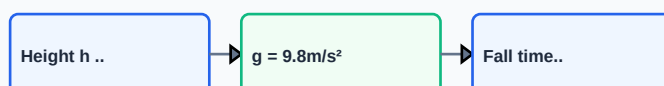
$$F = G * M * m / R^2 \quad F = (6.67 \times 10^{-11}) * (6 \times 10^{24}) * 1 / (6.4 \times 10^6)^2 \quad F = 40.02 \times 10^{13} / 4.096 \times 10^{13} = 9.77 \text{ Newtons (close to } 9.8 \text{ N).}$$

Example 2: An object weighs 60 N on Earth. What is its mass and weight on the Moon? (Moon gravity is 1/6th of Earth gravity).

Step-by-step Solution:

Mass on Earth = $W / g = 60 / 9.8 \approx 6.12 \text{ kg}$. Mass remains constant, so mass on Moon = 6.12 kg. Weight on Moon = Weight on Earth / 6 = $60 / 6 = 10 \text{ N}$.

WORKED: FREE FALL ACCELERATION AND VELOCITY



HOTS (Higher Order Thinking Skills) Challenge

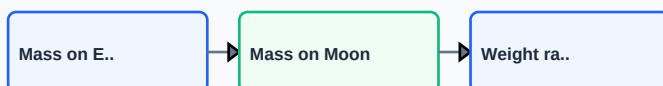
Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Why does acceleration due to gravity (g) vary at different latitudes on Earth, being maximum at the poles and minimum at the equator?

Analytical Solution Strategy:

The Earth is not a perfect sphere but an oblate spheroid. The polar radius is about 21 km shorter than the equatorial radius. Since $g = GM / R^2$, and R is smaller at the poles, g is larger. Rotation of the Earth also generates centrifugal forces that counter gravity slightly at the equator.

HOTS: GRAVITATIONAL ACCELERATION VARIATIONS AT POLES



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): State the universal law of gravitation and explain its significance.

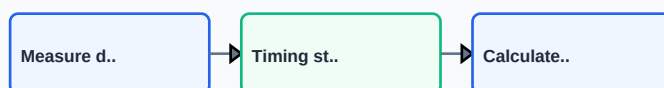
Q2 (2-Mark): Compare and contrast mass and weight.

Q3 (2-Mark): Explain how Kepler's three laws describe planetary motion.

Q4 (5-Mark): What is weightlessness? Explain the apparent weight of a person in a free-falling elevator.

Q5 (5-Mark): If the distance between two masses is doubled, how does the gravitational force change?

PRACTICE: GRAVITATIONAL CONSTANT DERIVATION SHEET



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Satellite Technology

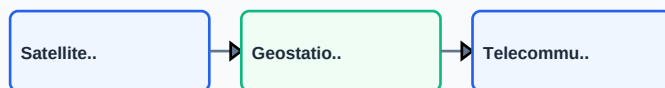
Orbital launch speeds and orbits of communications satellites are determined using gravitational balancing equations.

Application Case: Tide Modeling

High and low ocean tides are calculated based on the gravitational pull vectors of the Moon and the Sun on Earth's oceans.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

APPLICATION: ORBITAL MECHANICS IN SATELLITE POSITIONING



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

REVISION: GRAVITATION CONCEPTS LOOP

